



Interpretable radiomics method for predicting human papillomavirus status in oropharyngeal cancer using Bayesian networks

Oya Altinok^{a,b,*}, Albert Guvenis^a

^a Institute of Biomedical Engineering, Bogazici University, Istanbul, Turkey

^b Biomedical Engineering, Namik Kemal University, Tekirdag, Turkey

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ABSTRACT

Objectives: To develop a simple interpretable Bayesian Network (BN) to classify HPV status in patients with oropharyngeal cancer.

Methods: Two hundred forty-six patients, 216 of whom were HPV positive, were used in this study. We extracted 851 radiomics markers from patients' contrast-enhanced Computed Tomography (CT) images. Mens eX Machina (MXM) approach selected two most relevant predictors: sphericity and max2DDiameterRow. The area under the curve (AUC) demonstrated BN model performance in 30% of the data reserved for testing. A Support Vector Machine (SVM) based method was also implemented for comparison purposes.

Results: The Mens eX Machina (MXM) approach selected two most relevant predictors: sphericity and max2DDiameterRow. Areas under the Curves (AUC) were found 0.78 and 0.72 on the training and test data, respectively. When using support vector machine (SVM) and 25 features, the AUC was found 0.83 on the test data.

Conclusions: The straightforward structure and power of interpretability of our BN model will help clinicians make treatment decisions and enable the non-invasive detection of HPV status from contrast-enhanced CT images. Higher accuracy can be obtained using more complex structures at the expense of lower interpretability.

Advances in Knowledge: Radiomics is being studied lately as a simple imaging data based HPV status detection technique which can be an alternative to laboratory approaches. However, it generally lacks interpretability. This work demonstrated the feasibility of using Bayesian networks based radiomics for predicting HPV positivity in an interpretable way.

1. Introduction

Oropharyngeal cancer (OPC) is a type of head and neck cancer whose incidence has recently increased. Well-known risk factors for OPC are smoking, alcohol, oral hygiene, diet, and genetics. In 2007, the International Agency for Research on Cancer (IARC) accepted the human papillomavirus (HPV) infection as a risk factor for oropharyngeal cancer [1]. HPV-related cases in OPC have more than doubled over the past four decades [2], and today up to 63% of OPC are found to be HPV-infected [3]. More importantly, HPV-related OPC has a better prognosis. Studies have reported an increased success rate of response to treatment in patients with OPC who test positive for HPV [3–5]. Predicted life expectancy has also been higher in HPV-associated patients [3–5]. This may be because HPV-positive tumors are more radiosensitive [6] and, in turn, respond better to radiotherapy. In this regard, there is ongoing research on de-escalation studies [7], and several clinical trials [8,9]

have reported promising results of dose and toxicity reduction strategies. Therefore, accurately determining HPV status can not only help us obtain diagnostic and prognostic information but also help us design personalized treatment strategies for OPC patients.

Radiologists do not currently evaluate histopathological features such as tumor grade, lymphovascular invasion, perineural invasion, extracapsular invasion, and HPV status [10]. In addition, in some cases tumors are heterogeneous and difficult to biopsy. Quantitative image analysis methods (ie radiomics) of these features could be clinically useful. Thus, radiomics can aid treatment planning and improve treatment decisions.

HPV-infected cancers have been reported to be radiologically different from uninfected tumors [11–13]. Radiomics is a technique to extract meaningful information, i.e., imaging biomarkers, from medical images. Radiomics-based machine learning methods have been shown to predict overall survival, cancer recurrence, and metastasis in OPC.

* Corresponding author at: Institute of Biomedical Engineering, Bogazici University, Istanbul, Turkey.

E-mail address: oya.altinok@boun.edu.tr (O. Altinok).

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Thus, this suggests that profiling cancer by non-invasive techniques can be achieved. Accordingly, in recent years, radiomics biomarkers have been demonstrated to have the ability to identify HPV positivity in head and neck cancer [13,14]. One group [15] studied textural analysis of contrast-enhanced CT images of oropharyngeal cancer patients and showed statistically significant textural differences between HPV-positive and negative tumors. These methods have shown to have a high success in detecting the HPV status however interpretability has not received much attention. These models are black box types as suggested in [16–18]. A review work on general strategies that can be used for improving interpretability in deep learning based oncological radiomics is given in [19]. Interpretability of machine learning models is a requirement that serves to build confidence in actual clinical use. However, these models may yield lower accuracy. It is therefore important to use both types of models in a complementary way. In this study we address this unmet need and propose a Bayesian Network (BN) model for determining HPV status. BN is known to be an interpretable probability-based machine learning model [20].

If humans can easily understand the prediction of a machine learning model, the model is called interpretable. Mihaljević et al. [21] suggest that BN is a suitable approach for interpretable machine learning modelling. Simulatability, decomposability, and algorithmic transparency are the general characteristics of an interpretable model defined by Lipton [22], Arrieta et al. [17] and Murdoch et al. [23]. BN satisfies all these characteristics. Simulatability is how the user can simulate the model in his mind. (how a model works internally) How much the user can understand the internal workings of the model is algorithmic transparency. Since BN provides interactive graphical representation, it naturally satisfies transparency and simulatability. Decomposability, on the other hand, is an ability to explain every input parameter in the model. Further details can be found in [21].

Bayesian networks are directed acyclic graphs (DAG) that reveal probabilistic relationships between variables [24]. Nodes and arcs are two elements of DAG. Nodes represent radiomic biomarker variables while arcs connect nodes and show probabilistic associations between nodes. Regarding quantitative computations of the DAG, conditional probabilities are defined for each variable (node). In addition, DAG is a way of visual representation of the model. In terms of user-model interaction, visual representation is a straightforward explanation of the model [25]. Therefore, network analysis for predictive modeling provides interpretability to professionals seeking to understand the interrelationships of variables.

There are three ways to identify the structure of DAG. One way is expert knowledge, the second way is to learn from the data, and the third way is the combination of the first two. However, it is more challenging to learn DAG from data, as the complexity of computations and possible structures (DAGs) increases as the number of variables increases [26].

This study aims to analyze the value of CT-based radiomics using an interpretable and straightforward BN model for predicting human papillomavirus (HPV) in oropharyngeal cancer.

Here, we developed a straightforward BN structure to determine HPV positivity to guide medical practitioners in designing personalized therapy for patients with oropharyngeal cancer.

2. Materials and methods

Fig. 1 represents the schematic overview of our methodology. Figure S1 (Supplementary materials) shows the overall workflow. A Support Vector Machine (SVM) based method was also implemented for comparison purposes (Figure S2). The source code is available at <https://github.com/OA231/BN.git>.

2.1. Data set

Contrast-enhanced Computed Tomography (CT) images (segmentation labeled) of OPC were obtained from The Cancer Imaging Archive (TCIA) [27–29]. Gross primary tumor volume (GTVp) and gross nodal tumor volume (GTVn) are the regions of interest (ROI) segmented by experts manually, as described in [27]. In [30,31] only GTVp is considered for radiomic extraction. Therefore, since our concern is to determine HPV status, we also consider ROIs defined as GTVp in this dataset.

2.2. Image preprocessing

Some subjects were not included because ROIs associated with the images were not found. All patient images were resampled to $1 \text{ mm} \times 1 \text{ mm} \times 1 \text{ mm}$ spatial resolution to avoid varying voxel sizes [27] and make them comparable. The B-spline interpolation [32] method was used to find new values. Resampling and interpolation were done in 3D Slicer software (version 4.11.2) [33].

2.3. Feature extraction

Segmentation has been already performed by two expert oncologists as described in [28]. Feature extraction is converting medical images (ROI) into quantitative features. After image preprocessing, radiomics features were extracted by using 3D Slicer software (version 4.11.2) [33] with PyRadiomics [34] extension. Grey-level discretization was performed by setting the bin width to 10 to reduce heterogeneity [35] in the 3D slicer. The radiomic features of original images and wavelet transform images were extracted in 5 different categories: shape, intensity statistics, gray level co-occurrence matrix (GLCM), gray level run length matrix (GLRLM), and gray level size zone matrix (GLSZM). Full

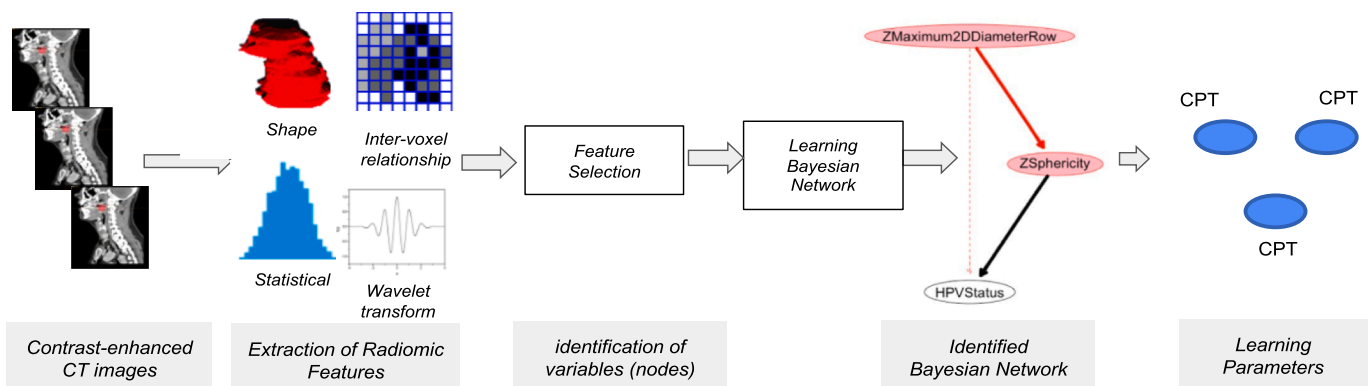


Fig. 1. Schematic view of the methodology. CPT depicts conditional probabilities between nodes.

descriptions of these radiomic categories can be found in [34].

2.4. Data preprocessing

Z-score normalization was applied to high dimensional data with 851 radiomics features. Continuous data were discretized into four levels using interval method of the to improve classification performance[36]. The dataset was split for 70% training and 30% testing. However, there was inequality in the number of HPV cases. The proportion of patients who are HPV negative was approximately 15%, and the rest were positive in the dataset. Therefore, due to the inequality in HPV cases, Random Over-Sampling Examples (ROSE)[37] in R[38] was applied to the training dataset to generate synthetic data. The test data was not subject to ROSE.

2.5. Feature selection

Two statistical methods were applied to obtain extracted data's most informative radiomics features. First, an independent *t*-test was performed between HPV-positive and negative categories in SPSS 27, yielding the 25 most valuable features. The data were considered significant with a *p*-value of < 0.05 .

Mihaljevic et al. [21] emphasizes that as the number of variables in the model increases for Bayesian Networks, the complexity of the model increases and its intelligibility decreases. Therefore, to identify the smallest number of features is essential to have simple structured BN model. To do so, a feature selection algorithm called Max-Min Parents and Children (MMPC) contained in a R package [39] was employed as a second step of feature selection.

MMPC is a constraint-based feature selection algorithm first introduced in [40]. Based on the given dataset, the main purpose of this algorithm is to determine the parent-child subset of the target variable (HPVStatus). For this, multiple conditional independence tests are performed repeatedly, and irrelevant and unnecessary variables which are not conditionally dependent on the target are eliminated accordingly. Consequently, the resulting set of variables (i.e., the parent-child set) is called the minimal subset of features for BN model.

Since HPVStatus is a two-level (positive and negative) binary variable, the MMPC algorithm used logistic regression for conditional independence tests. These tests were based on the likelihood ratio. Thus, in conditioning-independent tests, *p* values were checked whether the candidate variable was independent of the target variable (HPVStatus). Candidate variables that had no effect on the target variable were progressively excluded. The threshold for the *p* value was set to 0.01.

2.6. Learning the Bayesian network structure and parameters

We used R[38] package *bnlearn*[41] to learn BN structure with two predictors: Sphericity and Max2DDiameterRow. HPVStatus is a binary variable, which shows whether a patient has an HPV infection or not. We used the hill-climbing algorithm by means of a score-based structure learning approach, to learn the network structure. Then, parameters were learned using the *bn.fit* function. By doing so, conditional probabilities of nodes (variables) were obtained. Only modified training data was used to learn the structure and the parameters.

2.7. Cross-validation

5-fold cross-validation on the training dataset was performed to evaluate the performance of the predictive model. Training data was partitioned into five subsets with equal sizes. The prediction error for HPVStatus was set as a loss function. The resulting structure consists of the *test*, *fitted*, and *learning* elements. First, we predicted the values for test cases in each fold *k*. Next, the *fitted* element was converted to a *gRain* object, and then the *k*-th test subset was created.

3. Performance measurement of the predictive model

Training data. The area under the curve (AUC) evaluated model performance. We used the *caTools* package (Version: 1.18.2) to evaluate our model's performance.

Test data. The *gRain*[42] package was used to perform posterior inferences of the fitted BN. First, BN parameters were learned (on training data). Then, we used *test data* to predict HPV class. The area under the curve (AUC) evaluated model performance.

3.1. Measurement of arc strength

The relationship of the connected nodes was measured using the *arc.strength* function, and *criterion = X2* was set. The strength represents a probabilistic relationship of an arc of BN. It computes strength for each arc and keeps fixed the rest of the network structure. The criterion is a conditional independence test, and the strength measures a *p*-value. Therefore, more robust relationships are lower *p*-values and versa. Furthermore, we computed the strength of every possible arc. This function computes both frequencies of arcs presented after bootstrap samples and the direction of arcs.

3.2. Statistical analysis

Statistical analyses were done using IBM SPSS 27. Independent samples *t*-test analysis was performed. All tests were two-tailed.

4. Results

Patient selection diagram is shown in Fig. 2 and patient characteristics are given in Table 1.

As shown in Fig. 2, the TCIA dataset includes 412 OPC patients; but not all of them have HPV test results. One of the reasons why HPV status is not available to all patients may be that HPV testing for OPC is not part of standard care. Therefore, patients with unknown HPV (145) in the dataset were excluded from this study. In addition, images of 21 patients were excluded due to segmentation errors. After removing these data, 246 patients remained (Fig. 2). Example images can be seen in Fig. 6.

Two hundred sixteen patients were HPV positive, and the rest were negative. The patients were split %70 and %30 as training (172 patients) and testing (74 patients). Modified training data set included 115 HPV-positive patients and 135 HPV-negative patients. The final training data consisted of 250 patients with approximately equal HPV positive and negative patient rates.

First, by applying an independent sample *t*-test, 25 radiomic features were selected between HPV-positive and HPV-negative cases. Then, two radiomics features were selected by Mens eX Machina (MXM), a feature selection method in R. Sphericity and Max2DDiameterRow were identified after the feature selection process. These two radiomic biomarkers are shape features of the tumor. Fig. 3A shows that HPV-positive tumors were more spherical than HPV-negative tumors, and Max2DDiameterRow (Fig. 3B) was more significant in HPV-negative tumors. Our hypothesis was that these identified radiomic features, extracted from CT images, can predict HPV status noninvasively in oropharyngeal cancer patients.

Therefore, we developed a straightforward BN structure to determine HPV positivity (Fig. 4). In the BN structure, directed arcs show the casual relationship of nodes. Red lines represent the negative association between nodes, while the black line represents the positive association. The thickness of an arc represents the strength of a connection between nodes.

4.1. Analyzing of the network structure and its parameters

Identified continuous variables (predictors) were discretized using the *interval* method into four levels. Modified training data set were used

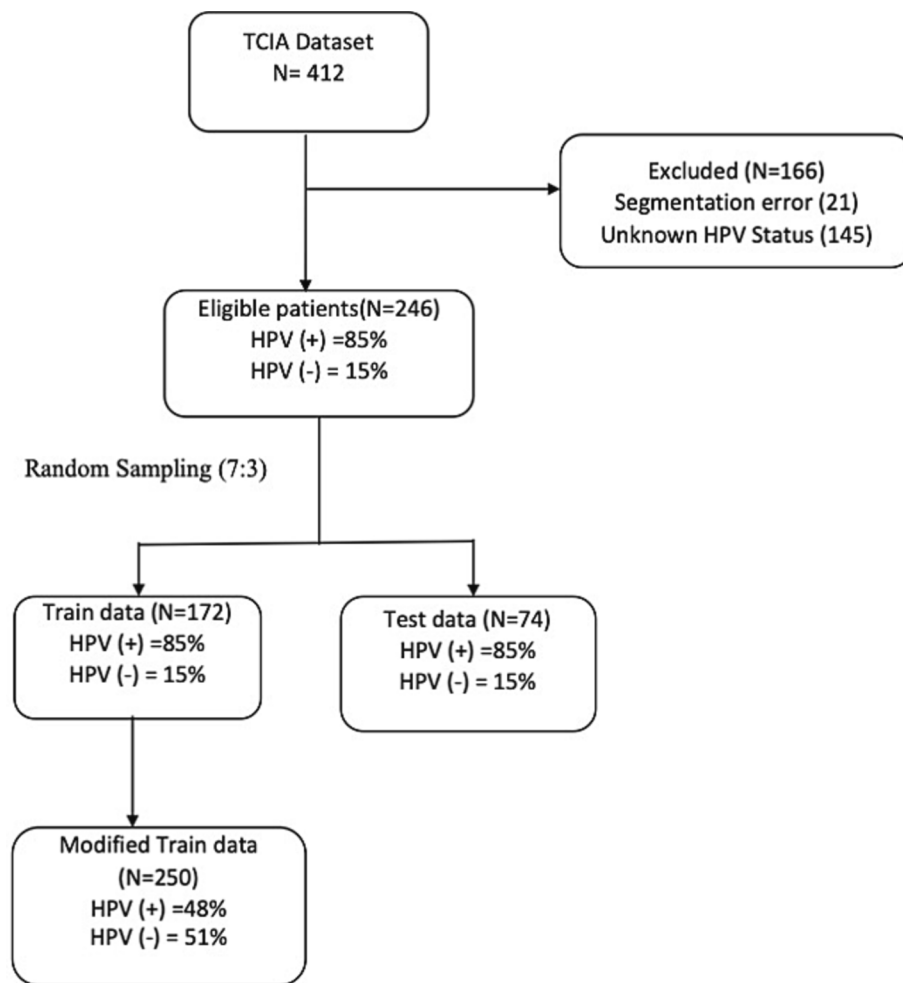


Fig. 2. This figure gives the patient selection diagram based on the data given in [26–28].

Table 1

This table has been created based on the data given in [27–29]. Patient characteristics of all included patients. Continuous variables are represented as means $\pm$ standard deviation.

Characteristics	Statistics (N = 246)
Age(mean) (yr)	58.3 $\pm$ 9.3
Sex (No.) (%)	
Female	44 (17.9%)
Male	202 (82.1%)
Smoking (No.) (%)	
Current	60 (24.4%)
Former	97 (39.4%)
Never	89 (36.2%)
HPV Status (No.) (%)	
Negative	36 (14.6%)
Positive	219 (85.4%)
T Category (No.) (%)	
1	63 (25.6%)
2	97 (39.4%)
3	50 (20.3%)
4	36 (14.6%)
N Category (No.) (%)	
0	23 (9.3%)
1	31 (12.6%)
2	186 (75.6%)
3	6 (2.4%)

to learn the BN structure and parameters. We obtained our predictive BN structure from the training dataset using a hill-climbing algorithm. The multinomial log-likelihood (loglik) score calculated BN. Sphericity and

Max2DDiameterRow are called the parents node of HPVStatus in the network. These parent nodes of the structure contain predictive parameters of the HPVStatus node.

We analyzed the arc strengths of the network (Fig. 4) to evaluate the probabilistic relationship of nodes (variables). The arc from Max2D-DiameterRow to Sphericity shows a robust negative association (p-value < 0.0001), while the arc from Max2DDiameterRow to HPVStatus has a weak association (p-value = 0.27). On the other hand, arc strength from Sphericity to HPV status is strong with p-value = 0.028.

After the BN structure was identified, the parameters of the network were learned. First, we fit parameters of the network using their bayes parameter estimation. Thus, the conditional probability tables for nodes and the class probabilities were determined (Table 2). This allows us to predict HPV probabilities for new observations.

4.2. Interpretability and explainability of the model

Class variable (HPV status) can be determined with maximum probability from the BN with joint probability distribution (JPD). JPD allows us to do probabilistic reasoning to estimate the value of a particular node given the values of other nodes. Sphericity and Max2-DiameterRow are two input nodes of our BN. To estimate a probability of outcome, we can set evidence for both nodes and query the probability of HPV being positive or negative. Table 2 shows the conditional probabilities table of our model that is learned from the modified training data. For example, if sphericity = [0.29, 0.42] and Max2D-DiameterRow = [12.16, 37.04], it is highly probably that a patient is

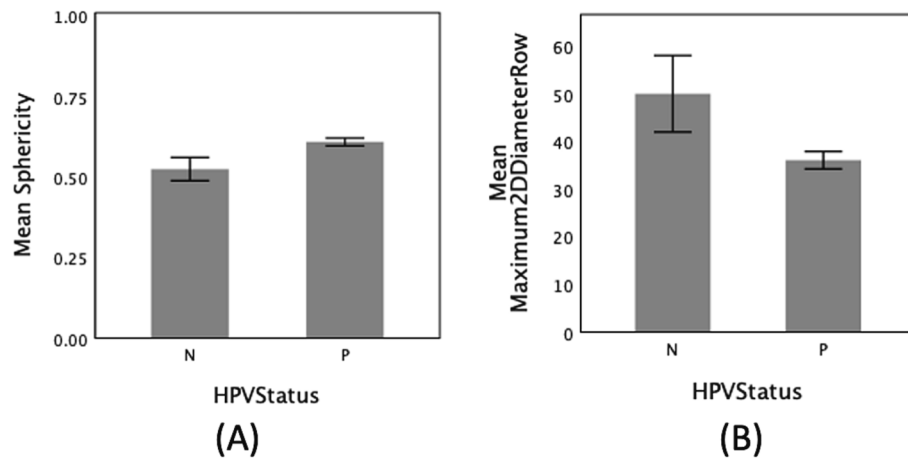


Fig. 3. Shows statistical significance between hpv positive and negative patients. p represents positive cases and n represents negative cases. (a) comparison of mean sphericity ($p < 0.001$), (B) Comparison of mean Max2DDiameterRow ($p < 0.001$).

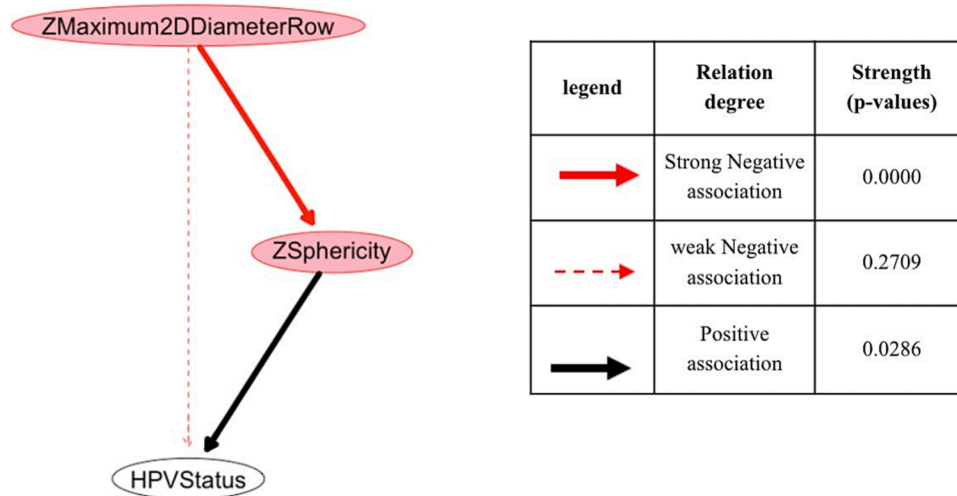


Fig. 4. Predictive BN model structure. Pink nodes represent predictor values, and HPVStatus node is a class variable that classifies HPV status. The directed arcs show the probabilistic dependencies between nodes. Strength measures a p-value of an arc that expresses probabilistic relationships. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

HPV positive (99.68%) (Table 2). In addition, the visual graph of the BN model (Fig. 4) can be considered the model's explanatory power. As explained in the previous result section (analyzing of the network structure), our simple BN model can visually explain the interrelationship of radiomic features.

4.3. The performance of the BN model on the training dataset

We computed the performance of our predictive BN model on the modified training dataset based on cross validation (methods). On the training dataset, the AUC of predictions is 0.73–0.90, with an average AUC of 0.78. ROC curve of BN with training dataset is shown in Fig. 5 (A).

4.4. The performance of the BN model on the test dataset

To evaluate our BN model, we measured its performance on the test dataset. Trained BN was used in this context. We did not change any parameters of trained BN, and testing data had no contribution to structure learning and parameter learning of the BN. The model performance tested on test dataset had an AUC of 0.72. ROC curve of BN with the testing dataset is illustrated in Fig. 5 (B).

5. Discussion

Understanding the degree of relationships between variables (nodes) is a crucial advantage over frequency-based machine learning methods. However, for many methods it is hard to understand how a model handles classification. Artificial neural networks (ANNs) and support vector machines (SVMs) are two well-known examples of black-box classifiers [43]. In addition, logistic regression is also considered a black box in a very recent study [16]. Cynthia Rudin, in her article [18], warns about the use of “black-box machine learning models” for risky decisions and emphasizes the need for a shift to “interpretable models”. Mihaljević et al. [21] suggest that BN is a suitable approach for interpretable machine learning modelling.

In terms of interpretability of a model, transparency is used as a counter term to the black box[22]. Barredo Arrieta et al. [17] evaluates different machine learning methods under the concept of transparency. Linear and logistic regression models are considered transparent, but they require post-hoc analyses to make them interpretable, and their interpretability may depend on who will be the user [17]. However, classifiers should be easy to understand, as they are intended to assist professionals in diagnosis or prognosis. Table 3 compared several machine learning models for HPV status prediction in CT of head and neck

Table 2

The table of conditional probabilities P (HVPStatus | Sphericity, Max2DDiameterRow) The data were discretized into four levels. Sphericity (0.596 ± 0.967) and Max2DDiameterRow (38.23 ± 16.31), here, are actual values for illustration purposes. They are represented as means $\pm$ standard deviation.

Sphericity = [0.29, 0.42]				
Maximum2DDiameterRow	[12.16, 37.04]	(37.04, 61.91]	(61.91, 86.79]	(86.79, 111.66]
HPV(Positive)	0.997	0.001	0.0016	0.001
HPV(Negative)	0.003	0.999	0.998	0.999
Sphericity = (0.42, 0.55]				
Maximum2DDiameterRow	[12.16, 37.04]	(37.04, 61.91]	(61.91, 86.79]	(86.79, 111.66]
HPV(Positive)	0.333	0.356	0.235	0.0006
HPV(Negative)	0.667	0.644	0.765	0.999
Sphericity = (0.55, 0.68]				
Maximum2DDiameterRow	[12.16, 37.04]	(37.04, 61.91]	(61.91, 86.79]	(86.79, 111.66]
HPV(Positive)	0.776	0.694	0.998	0.500
HPV(Negative)	0.224	0.306	0.0016	0.500
Sphericity = (0.68, 0.81]				
Maximum2DDiameterRow	[12.16, 37.04]	(37.04, 61.91]	(61.91, 86.79]	(86.79, 111.66]
HPV(Positive)	0.757	0.445	0.500	0.500
HPV(Negative)	0.242	0.556	0.500	0.500

cancer patients. Although many of these result in better performance (AUC) than our model, the problem of interpretability arises as suggested in [16,17].

With regard to models of Bager-Ebadian et. al and Bogowicz et. al, both models result in better performance (AUC) than our BN model; however, the problem of their model is lack of interpretability as suggested in [16–18]. For comparison purposes, the SVM model was

generated, and the AUC was resulted in 0.83, which is again higher than our BN model. This might be because that the SVM model used the 25 most valuable features, while the BN model used only 2 features. Mihaljevic et al. [21] emphasizes that as the number of variables in the model increases for Bayesian Networks, the complexity of the model increases and its intelligibility decreases. With this in mind, the number of features in our BN model has been deliberately limited to the minimum set of features, and to do this we have implemented an additional feature selection method [24] for the BN. This shows that higher accuracy can be achieved by using more complex structures (more features and black box modelling) at the expense of lower interpretability. The simultaneous use of both types of models can be considered and investigated.

This dataset was used for Medical Image Computing and Computer Assisted Intervention (MICCAI) grand challenge in 2016. Yu et al. [44] were the winner of the competition, and their results were published in [44]. Their model (Table 3) performance resulted in AUC (0.753), which is a comparable performance to our BN model AUC (0.78) on their training data. However, their model's performance (AUC = 0.915) on test data was surprisingly and considerably higher than the performance on the train data. Our model, together with its AUC (0.78 and 0.72 in training and test data, respectively) and interpretability capacity (see Table 2), demonstrates a robust and stable model for predicting HPV status although the accuracy is lower. The obtained results can be used as a basis for undertaking further studies with larger and more balanced data.

Apart from measuring the model's performance (AUC), BN has a strong interpretation (probabilistic reasoning) capacity of the model. For example, providing one or more evidence to the model allows the probability of having HPV to be calculated (Table 2). In the clinical aspect, this could allow medical practitioners to easily understand the

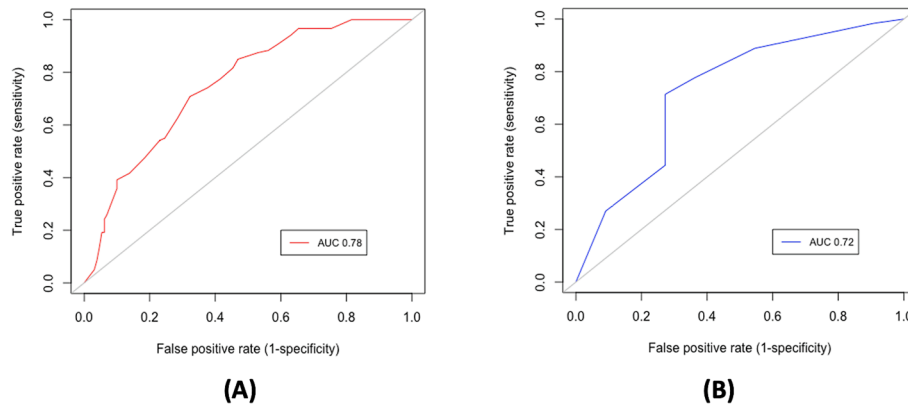


Fig. 5. (A) ROC curve of BN as shown in Fig. 3 based on cross validation with the modified training dataset. (B) ROC curve of BN with the test dataset.

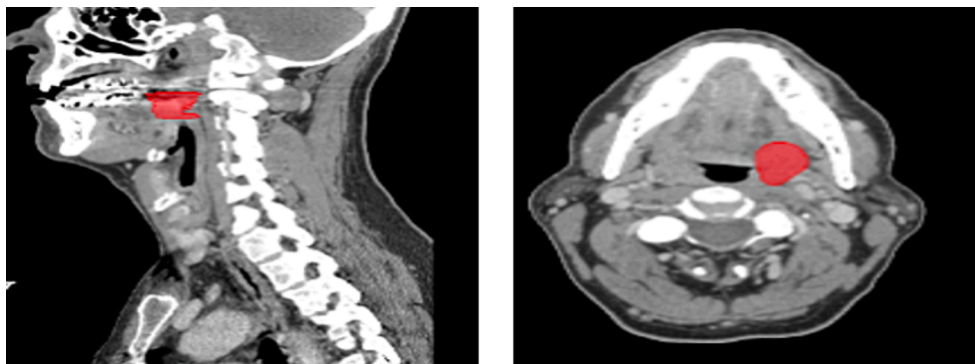


Fig. 6. Example CT slices. Red label depicts gross primary tumor volume (GTVp). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3

Comparison of the proposed method with machine learning models for HPV status prediction in CT of head and neck cancer patients. N represents the total number of features used in the model.

	Number of patients	Classifier	Performance (AUC)		Features
			Train	Test	
Yu et al. [44]	315 oropharynx cancer patients	Logistic regression	0.753	0.915	(N = 2 features) Shape Features: MeanBreadth & SphericalDisproportion
Bagher-Ebadian et al. [45]	187 oropharyngeal cancer patients	General Linear Model	0.849	0.869	(N = 12 features) GLCM – Contrast, Energy, Entrophy Shape Features -MeanBreadth & SphericalDisproportion DOST- Energy1,2,3,4,6,7,8,9
Bogowicz et al. [31]	149 HNSCC -Oropharynx (100) -Other sites (49)	Multivariable logistic regression	0.85	0.78	(N = 4 features) LLL standard deviation, LLL small zone high gray-level emphasis, HHL difference entropy, Coefficient of variation
Our work	246 oropharynx cancer patients	Support Vector Machine (SVM)		0.83	(N = 25 features)
Our work	246 oropharynx cancer patients	Bayesian Network (BN)	0.78	0.72	(N = 2 features) Sphericity, Max2DDiameterRow

results of the model and design personalized therapies.

Regarding to interpretability, our BN can predict HPV status and reveal probabilistic associations between variables (nodes). For example, arrows entering node “HPVStatus” in Fig. 4 explain the probability of HPV positivity by the extracted CT based shape characteristics. We identified two prominent variables: sphericity and Max2DDiameterRow. Sphericity ranges from zero to one and calculates the volume of the tumor compared to a sphere [46]. The closer the tumor volume is to zero, the farther from spherical and closer to one,

the more spherical it is. Sphericity has been demonstrated in the study [47] as a promising radiomic biomarker extracted from MRI images of breast cancer patients to predict treatment outcomes. Max2DDiameterRow, on the other hand, is defined as the largest binary Euclidean distance on the tumor surface [34]. The largest Max2DDiameterRow implies the complexity of tumor shape since tumor complexity has been shown to be associated with radiomic shape features [48]. One group [49] studied differences in imaging characteristics and showed that HPV-related tumors have well-defined borders in oropharyngeal cancers while HPV-negative tumors have ill-defined borders. In this context, this explains why HPV-positive tumors are more spherical and less complex.

HPV-positive oropharyngeal tumors are biologically [50] and radiologically [11] different and more manageable [51] compared to non-positive patients with high overall survival rates [3]. Maniakas et. al [52] surveyed North America and found that 67% of patients had undergone HPV laboratory testing; however, only 58.3% of these tests were utilized to treat head and neck cancer patients. Therefore, it is necessary that HPV-positive patients be identified easily to apply the appropriate treatment. While invasive techniques are common ways to test, HPV testing may not be performed in certain cases [52] CT-based radiomic models can provide physicians with an alternative. In addition, the BN model can provide a HPV positive status probability for for a new patient image. For instance, in Table 2, we calculated probabilities of a scenario that a patient has Max2DDiameterRow = [61.91, 86.79] (3rd level interval in Table 2) and Sphericity = [0.29, 0.42] (smallest interval), then the patient would have 0.16% of being HPV-positive. On the other hand, as tumor becomes more spherical, for instance [0.55, 0.68] (3rd level interval), with same max2DDiameterRow level, likelihood of being HPV-positive increases (99.8%).

5.1. Limitations and future directions

Our study has several limitations. First, the number of HPV-positive cases was considerably higher than the number of HPV-negative cases. This causes what is known as the class imbalance in machine learning. In addition, further investigation can be done on a larger and independent

external validation set to evaluate the predictive power of our BN model. Applying our model to larger datasets with more class homogeneity will be considered as future work since there is no optimal rule to solve the imbalance problem [53].

Second, Variations in CT scanners and protocols are currently a major limitation. Reiazi et. al. [54] recently published a review of how variations in CT manufacturers affect the robustness of radiomic features extracted from these CTs. They found that variations in the scanners have an important effect on the radiomic features. This causes the model’s performance to degrade. Another study [30] evaluated how radiomics from different CT scanners affected the prediction of HPV Status in OPC patients and found that the accuracy of the models in predicting HPV status differed between different CT scanners. New algorithms are now available for the harmonization of CT images [55]. Future work is needed to study if these corrections have a positive effect on the accuracy of predicting the HPV status.

Third, we used a minimal subset of variables to learn the structure of our BN model. Since BN is based on Bayesian statistics, it can be easily updated with new evidence or expert knowledge (new features). Therefore, our model can be expanded with expert knowledge to identify clinical predictors that can improve our model reliability.

In addition, the ethnicity of patients was not available in the dataset that was used in this study. However, research [56] have revealed that the race is an important factor for survival in HPV-negative OPC, but for HPV-positive cases, ethnicity was found to be not associated with survival. Future studies are needed to study whether ethnicity will be an important variable for the BN model.

It is well known that accuracy generally suffers when introducing interpretability [17]. The accuracy however may still be improved in future work by addressing the limitations indicated above. The use of a more balanced and harmonized dataset as well as clinical predictors and ethnicity as additional features may help to improve the accuracy in future work.

Some histopathological characteristics such as lymphovascular invasion, perineural invasion and HPV status are not currently evaluated by radiologists. Thus, our BN model may be clinically useful in assessing these these characteristics noninvasively as an alternative method. Furthermore, our methodology may be used and evaluated for other types of cancer.

6. Conclusion

Our study shows that the Bayesian network can predict HPV status from radiomic markers extracted from contrast-enhanced CT images. Additionally, the model can demonstrate probabilistic relationships between predictors (nodes). Unlike “black-box machine learning

models”, BN is a transparent model which makes it possible to interpret the results. This is clinically very valuable.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ejmp.2023.102671>.

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